USER'S MANUAL

WASTE-TO-WORTH: ENVIRONMENTALLY TRANSFORMATIVE USE OF IMAGE RECOGNITION, AND ARTIFICIAL INTELLIGENCE

October, 2025

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1.0 GENERAL INFORMATION

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1.1 System Overview

Waste to Worth is a comprehensive mobile recycling application designed to make waste management and recycling ridiculously easy for users. The system combines artificial intelligence, computer vision, and gamification to transform how people interact with waste materials and learn about sustainable practices.

Major Functions Performed by the System:

- **AI-Powered Material Recognition** - Uses computer vision and machine learning to identify waste materials through camera scanning
- **Climate-Aware Disposal Guidance** - Provides intelligent, location-specific disposal instructions using Google's Gemini AI
- **Gamified Learning Experience** - Features quest system, points, achievements, and progress tracking to encourage sustainable behaviors
- **Community Engagement** - Social platform for sharing recycling projects, tips, and experiences
- **Project-Based Learning** - Step-by-step upcycling and recycling project tutorials
- **Location-Based Services** - Automatic climate detection and location-aware recommendations
- **Progress Tracking** - Comprehensive history of scanned materials and completed projects

System Architecture:

The system follows a client-server architecture with the following components:

- **Frontend** - React Native mobile application with Expo framework, supporting iOS, Android, and web platforms
- **Backend** - Python Flask REST API server providing AI services, data management, and business logic
- **Database** - Google Firestore cloud database for user data, materials, projects, and community content
- **Services** - Google Gemini AI integration for intelligent disposal step generation
- **Image Storage** - Firebase Storage and Hostinger web hosting for media management
- **Authentication** - Firebase Authentication with login support

User Access Mode:

The system provides a graphical user interface (GUI) optimized for mobile devices with:

- Touch-based navigation and interactions
- Camera integration for material scanning
- Responsive design for various screen sizes

1.0 General Information

Responsible Organization:

San Beda College Alabang

Bachelor of Science in Information Technology

Principal Researcher: Yuri Andrei B. Morrison

Co-Researcher/s: Kristine Joie B. Mejia; Kylee A. Superficial

Bachelor of Science in Information Technology

General Description:

Waste to Worth transforms the traditional approach to waste management by making it engaging, educational, and accessible. Users can scan any waste item using their device's camera, receive instant AI-powered identification, and get personalized disposal instructions based on their local climate conditions. The app gamifies the recycling experience through quests, points, and achievements while fostering a community of environmentally conscious users.

System Environment or Special Conditions:

- **Platform Requirements** - Android 8+
- **Network Requirements** - Internet connection required for AI services and data synchronization
- **Permissions** - Camera access for material scanning, location access for climate-aware recommendations
- **Storage** - Local device storage for offline functionality and cached data
- **Climate Integration** - Automatic detection of tropical climate conditions for specialized disposal guidance

1.2 Project References

The following references were used in the preparation of this document, listed in order of importance to the end user:

- **Waste to Worth Application Source Code** - Primary system documentation and implementation details
- **Google Gemini AI Integration Summary** - AI-powered disposal guidance system documentation
- **Quest Generation System Documentation** - Gamification and achievement system specifications
- **Firebase Configuration and Setup** - Authentication and database integration details
- **Expo React Native Framework Documentation** - Mobile application development platform
- **Flask Python Backend API Documentation** - Server-side functionality and endpoints

- **Unsplash API Integration** - Image sourcing and management system

1.3 Authorized Use Permission

This system and all associated software, data, reports, and documents are protected by intellectual property rights.

Users are strictly prohibited from:

- Making unauthorized copies of the application software
- Redistributing or sharing user account credentials
- Attempting to reverse engineer or decompile the application
- Using the system for commercial purposes without explicit written permission
- Accessing or attempting to access other users' private data
- Modifying or tampering with the application's core functionality

Data Usage and Privacy:

- User-generated content (scans, posts, comments) remains the property of the individual users
- The system collects location and usage data to provide personalized services
- All data is stored securely using industry-standard encryption and security measures
- Users retain the right to delete their account and associated data at any time

Copy Permissions:

If you need to obtain permission for specific use cases, please contact the development team through the official support channels listed in section 1.4.

1.4 Points of Contact

Primary Contact: Yuri Andrei B. Morrison
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1.5 Acronyms and Abbreviations

The following acronyms and abbreviations are used throughout this document:

- **AI** - Artificial Intelligence
- **API** - Application Programming Interface

- **GUI** - Graphical User Interface
- **GPS** - Global Positioning System
- **HTTP** - HyperText Transfer Protocol
- **JSON** - JavaScript Object Notation
- **REST** - Representational State Transfer
- **SDK** - Software Development Kit
- **UI** - User Interface
- **UX** - User Experience
- **URL** - Uniform Resource Locator
- **QR** - Quick Response (code)
- **PNG** - Portable Network Graphics
- **JPEG** - Joint Photographic Experts Group
- **GIF** - Graphics Interchange Format
- **WebP** - Web Picture Format
- **MB** - Megabyte
- **GB** - Gigabyte
- **RAM** - Random Access Memory
- **CPU** - Central Processing Unit
- **OS** - Operating System
- **iOS** - iPhone Operating System
- **Android** - Google's mobile operating system
- **Web** - World Wide Web platform
- **Firebase** - Google's mobile and web application development platform
- **Firestore** - Google's NoSQL document database
- **Expo** - React Native development platform
- **Flask** - Python web framework
- **Gemini** - Google's AI model for content generation
- **Unsplash** - Stock photography platform
- **Hostinger** - Web hosting service provider
- **EULA** - End User License Agreement
- **GDPR** - General Data Protection Regulation
- **SSL** - Secure Sockets Layer
- **TLS** - Transport Layer Security
- **OAuth** - Open Authorization protocol
- **JWT** - JSON Web Token
- **CORS** - Cross-Origin Resource Sharing
- **MVP** - Minimum Viable Product
- **QA** - Quality Assurance
- **UAT** - User Acceptance Testing
- **SLA** - Service Level Agreement
- **FAQ** - Frequently Asked Questions
- **TOS** - Terms of Service
- **PP** - Privacy Policy

2.0 SYSTEM SUMMARY

2.0 SYSTEM SUMMARY

Waste to Worth is an intelligent recycling companion that transforms how people interact with waste materials in their daily lives. The system serves as a comprehensive digital assistant for environmental sustainability, helping users make informed decisions about waste disposal while learning about recycling practices through an engaging, gamified experience.The system supports users in their environmental journey by providing instant material identification, personalized disposal guidance, and educational content. For staff and administrators, the system offers comprehensive analytics, user engagement metrics, and content management capabilities to continuously improve the platform's effectiveness in promoting sustainable behaviors.

Users can seamlessly scan waste items using their mobile device's camera, receive immediate AI-powered identification and disposal instructions, track their environmental impact through points and achievements, and connect with a community of like-minded individuals. The system adapts to local climate conditions and provides location-specific recommendations, making it relevant and practical for users worldwide.

2.1 Hardware and Software Specifications

The Waste to Worth system operates across multiple platforms with varying hardware requirements and software specifications:

Mobile Device Requirements:

- Operating System: Android 8.0 (API level 26) or later
- Hardware: Devices with ARM64 or x86_64 architecture
- Storage: Minimum 100MB available space for app installation
- Memory: 2GB RAM recommended for optimal performance
- Camera: Rear-facing camera with autofocus capability
- Network: Wi-Fi or cellular data connection for AI services

Input Devices:

- Touch screen for navigation and interaction
- Device camera for material scanning and image capture
- Keyboard for text input in community features
- GPS/location services for climate-aware recommendations

Output Devices:

- Display screen for visual interface and results
- Audio speakers for notifications and feedback
- Haptic feedback for tactile responses
- Vibration for alerts and confirmations

2.2 User Access Levels

Registered Users:

- **Access Level** - Full application functionality with personal account
- **Capabilities** - Complete material scanning, quest participation, community engagement, progress tracking
- **Data Access** - Personal scan history, achievement records, profile information
- **Community Features** - Can create posts, comment, like content, share projects
- **Storage** - Scan history and project data
- **Personalization** - Climate-aware recommendations, customized quest suggestions

2.3 Contingencies and Alternate Modes of Operation

Waste to Worth implements comprehensive contingency planning to ensure continuous operation and minimal disruption to user experience:

AI Service Disruptions:

- Fallback to predefined disposal instructions database
- Material-specific guidance based on cached knowledge
- Community-sourced disposal tips as backup
- User Impact: Less personalized but still accurate disposal guidance

Database and Storage Failures

Firestore Database Issues:

- Local storage backup for critical user data
- Read-only mode with cached content
- Emergency data recovery procedures
- User Impact: Temporary inability to save new data, existing data remains accessible

Media Storage Problems:

- Alternative image hosting services
- Local device storage for critical images
- Compressed image formats for reduced storage requirements
- User Impact: Some images may not load, but core functionality continues

3.0 GETTING STARTED

3.0 GETTING STARTED

This section provides a general walkthrough of the system from initiation through exit. This section provides a comprehensive walkthrough of the Waste to Worth from initial access through proper exit procedures. The information is organized to help users understand the logical flow and sequence of system operations, enabling efficient navigation and optimal use of all available features.

The system follows an intuitive user journey that begins with account creation or login, progresses through the main application interface, and concludes with proper session termination. Each step is designed to guide users toward successful waste identification, disposal guidance, and community engagement while maintaining security and data integrity.

3.1 Logging On

Accessing the Waste to Worth system requires either creating a new account or logging into an existing account. The system supports multiple authentication methods to accommodate different user preferences and security requirements.

New User Registration:

- Launch the Waste to Worth application on your mobile device or web browser
- Select the "Sign Up" option from the initial login screen
- Choose your preferred registration method:
    - Email and password creation
    - Google account integration
    - Facebook account integration
- Complete the required information fields:
    - Email address (for email registration)
    - Secure password
- Accept the End User License Agreement (EULA) and Privacy Policy
- Verify your email address through the confirmation link sent to your registered email
- Complete the initial profile setup and onboarding process

User ID Assignment:

- System automatically generates a unique user identifier upon successful registration
- User ID is permanent and cannot be changed by the user
- Display name can be customized in profile settings

Existing User Login:

- Open the Waste to Worth application
- Enter your registered email address in the email field
- Enter your password in the password field
- Tap "Sign In" to authenticate
- System validates credentials and grants access to main application interface

3.2 System Menu

Upon successful login, users are presented with the main application interface featuring a bottom navigation bar with five primary sections. Each section provides access to specific system functions and features.

Main Navigation Structure:

The system menu consists of five primary sections accessible through the bottom navigation bar:

- **Home** - Dashboard and user overview
- **History** - Scan history and material tracking
- **Scan** - Camera interface for material identification
- **Community** - Social features and content sharing
- **Quests** - Gamification and achievement system

Primary Navigation:

- Tap any icon in the bottom navigation bar to switch between main sections
- Current section is highlighted with primary color and active state

Secondary Navigation:

- Settings menu accessible from gear icon in top-right corner of most screens
- Back button functionality for nested screens
- Breadcrumb navigation for complex workflows

3.2.1 Home Dashboard

The Home Dashboard serves as the central hub for user activity and system overview. This function provides a comprehensive view of user progress, recent activities, and quick access to primary system features.

Menu Pathway: Main navigation → Home tab (default landing screen)

Function Description:

- Displays user profile summary with achievement level and progress
- Shows recent scanned materials with thumbnail images
- Provides access to current active projects

Navigation Path:

- User logs into system
- Automatically redirected to Home Dashboard
- Access additional features through dashboard cards and buttons
- Navigate to other sections using bottom navigation bar

3.2.2 Material Scanning

The Material Scanning function is the core feature of the Waste to Worth system, enabling users to identify waste materials through camera-based image recognition and receive personalized disposal guidance.

Menu Pathway: Main navigation → Scan tab

Function Description:

- Real-time camera interface for material capture
- AI-powered material identification and classification
- Climate-aware disposal instruction generation
- Integration with quest system for progress tracking
- Automatic location detection for personalized recommendations

Navigation Path:

- Tap Scan icon in bottom navigation bar
- Grant camera permissions if not previously authorized
- Position waste item within camera frame
- Tap capture button to take photo
- System processes image and displays results
- Navigate to material detail page for comprehensive information

3.2.3 Scan History

The Scan History function provides users with comprehensive tracking of all materials they have scanned, including detailed information about disposal methods, project suggestions, and environmental impact.

Menu Pathway: Main navigation → History tab

Function Description:

- Chronological list of all scanned materials
- Detailed material information and disposal instructions
- Project suggestions based on scanned materials
- Search and filter capabilities for material history
- Export functionality for personal records

Navigation Path:

- Tap History icon in bottom navigation bar

- View chronological list of scanned materials
- Tap any material entry for detailed information
- Use search bar to find specific materials
- Access project suggestions from material details

3.2.4 Community Features

The Community function enables users to share their recycling experiences, learn from others, and participate in a supportive network of environmentally conscious individuals.

Menu Pathway: Main navigation → Community tab

Function Description:

- Create and share posts about recycling projects
- Comment on and like community content
- Upload photos of completed projects
- Access community-driven disposal tips and advice

Navigation Path:

- Tap Community icon in bottom navigation bar
- View community feed with recent posts
- Tap "+" button to create new post
- Select photos and add description
- Publish post to community feed
- Interact with other users' content through likes and comments

3.2.5 Quest System

The Quest System provides gamified challenges and achievements to encourage continued engagement with sustainable practices and system features.

Menu Pathway: Main navigation → Quests tab

Function Description:

- Daily, weekly, and monthly challenges
- Achievement badges and progress tracking
- Point system for completed activities
- Leaderboards and community competitions
- Personalized quest recommendations

Navigation Path:

- Tap Quests icon in bottom navigation bar
- View available quests and challenges
- Select quest to view requirements and rewards
- Complete quest activities through normal app usage

- Claim rewards and track progress
- View achievement gallery and statistics

3.3 Changing User ID and Password

User ID Characteristics:

- User IDs are system-generated and permanent
- Cannot be changed by users for security and data integrity
- Used for internal system operations and data association

Display Name Changes:

- Navigate to Settings → Profile Settings
- Select "Edit Profile" option
- Modify first name and last name fields
- Save changes to update display name
- Changes reflect immediately across all system interfaces

3.4 Exit System

Standard Logout Process:

- Navigate to Settings menu
- Select "Logout" option
- Confirm logout action when prompted
- System clears session data and returns to login screen
- User data is automatically saved and synchronized

4.0 USING THE SYSTEM

4.0 USING THE SYSTEM

This section provides a detailed description of system functions. Each function should be under a separate section header, 4.1 - 4.x, and should correspond sequentially to the system functions (menu items) listed in subsections 3.2.1 - 3.2.x.

4.1 Home Dashboard

The Home Dashboard serves as the central hub for user activity and system overview. This function provides a comprehensive view of user progress, recent activities, and quick access to primary system features.

The Home Dashboard displays user profile summary with achievement level and progress, shows recent scanned materials with thumbnail images, provides access to current active projects, displays quest progress and available challenges, and offers quick access to scanning functionality.

4.1.3 Active Projects Overview

This sub-function provides an overview of currently active recycling and upcycling projects that the user is working on.

Function Description:

- Shows projects currently in progress
- Displays project completion percentage
- Provides quick access to project details
- Shows next steps for each active project
- Allows users to mark projects as completed

4.2 Material Scanning

The Material Scanning function is the core feature of the Waste to Worth system, enabling users to identify waste materials through camera-based image recognition and receive personalized disposal guidance.

The Material Scanning function provides real-time camera interface for material capture, AI-powered material identification and classification, climate-aware disposal instruction generation, integration with quest system for progress tracking, and automatic location detection for personalized recommendations.

4.2.1 Camera Interface

This sub-function provides the camera interface for capturing images of waste materials for identification.

Function Description:
- Real-time camera preview with focus indicators

- Capture button for taking photos
- Flash control for low-light conditions
- Camera roll access for selecting existing photos
- Image quality optimization for AI processing

4.2.2 AI Material Recognition

This sub-function uses computer vision and machine learning to identify materials from captured images.

Function Description:
- Processes captured images using AI algorithms
- Identifies material type and composition
- Provides confidence scores for identification accuracy
- Offers alternative material suggestions if primary identification is uncertain
- Stores identification results for future reference

4.3 Scan History

The Scan History function provides users with comprehensive tracking of all materials they have scanned, including detailed information about disposal methods and project suggestions.

The Scan History function provides a chronological list of all scanned materials, detailed material information and disposal instructions, and project suggestions based on scanned materials.

4.4 Community Features

The Community function enables users to share their recycling experiences, learn from others, and participate in a supportive network of environmentally conscious individuals.

The Community function enables users to create and share posts about recycling projects, comment on and like community content, upload photos of completed projects, and interact with other users' content.

4.5 Quest System

The Quest System provides gamified challenges and achievements to encourage continued engagement with sustainable practices and system features.

The Quest System interface presents a gamified experience that encourages users to engage with sustainable practices through challenges and achievements. The main screen displays available quests organized by category, with progress tracking and reward information. Users can view their achievement progress, track progress toward various challenges, and see their current level and points. The system automatically tracks user activities and awards points and achievements for completed actions.

4.5.1 Quest Categories

This sub-function organizes quests into different categories based on activity type and difficulty.

Function Description:

- Scanning quests for material identification
- Community quests for social engagement
- Recycling quests for disposal activities
- Upcycling quests for creative projects
- Profile quests for account completion

4.6 Special Instructions for Error Correction

This section describes all recovery and error correction procedures, including error conditions that may be generated and corrective actions that may need to be taken.

Camera Permission Errors:

Error: "Camera access denied" or "Permission required"

Cause: User has not granted camera permissions

Corrective Action:

- Go to device Settings
- Navigate to App Permissions
- Enable Camera access for Waste to Worth
- Return to app and retry scanning

Network Connection Errors:

Error: "Network error" or "Unable to connect"

Cause: Poor internet connection or server unavailable

Corrective Action:

- Check internet connection
- Try switching between WiFi and mobile data
- Wait a few moments and retry
- Contact support if problem persists

Image Processing Errors:

Error: "Unable to identify material" or "Processing failed"

Cause: Poor image quality or unsupported material

Corrective Action:

- Ensure good lighting conditions
- Position material clearly in camera frame
- Avoid blurry or dark images
- Try scanning from different angles
- Ensure material is clearly visible

Authentication Errors:

Error: "Login failed" or "Session expired"

Cause: Invalid credentials or expired session

Corrective Action:

- Verify username and password
- Try logging out and back in
- Reset password if necessary
- Contact support for account issues

4.7 Caveats and Exceptions

Scan History Preservation:

- Scan history is automatically saved to the cloud database
- Users must have an active internet connection for data synchronization
- Offline scans will be uploaded when connection is restored
- Manual backup is not required but recommended for important data

Quest Progress:

- Quest progress is tracked automatically in real-time
- Users must complete quest activities while connected to the internet
- Progress may be delayed if connection is poor
- Quest completions are saved immediately upon achievement

Material Scanning:

- Camera permissions must be granted for scanning to work
- Good lighting conditions are required for accurate material identification
- Material must be clearly visible and well-positioned in camera frame
- Internet connection is required for AI processing and results

4.8 Input Procedures and Expected Output

This section provides detailed instructions describing the procedures the user will need to follow to use the system, including input formats, preparation rules, and output procedures.

Job Request Forms and Control Statements:

- Frequency: On-demand, as needed by user
- Reason: To access system functions and perform material scanning, community interaction, and quest activities
- Origin: User-initiated through mobile device interface
- Medium: Touch-based input through mobile device screen

Input Preparation Rules:

- Ensure device has adequate battery life for intended usage
- Verify internet connection is available for full functionality
- Grant necessary permissions (camera, location) when prompted

Expected Output:

- Purpose: Provide access to system functions and display results
- Frequency: Real-time response to user interactions
- Options: Various output formats including text, images, and interactive elements
- Media: Mobile device screen display

Codes and Abbreviations Used in System Output

Material Identification Codes:

- MAT_: Material type prefix (e.g., MAT_PLASTIC, MAT_GLASS)
- TRAIT_: Material trait codes (e.g., TRAIT_CLEAR, TRAIT_RIGID)
- DISPOSAL_: Disposal method codes (e.g., DISPOSAL_RECYCLE, DISPOSAL_COMPOST)

Quest System Codes:

- QUEST_: Quest type prefix (e.g., QUEST_SCAN, QUEST_COMMUNITY)

- ACHIEVEMENT_: Achievement level codes (e.g., ACHIEVEMENT_BRONZE, ACHIEVEMENT_GOLD)
- POINTS_: Point value indicators (e.g., POINTS_10, POINTS_50)

Community Codes:

- POST_: Post type codes (e.g., POST_PROJECT, POST_TIP)
- INTERACTION_: Interaction type codes (e.g., INTERACTION_LIKE, INTERACTION_COMMENT)

System Status Codes:

- STATUS_: System status indicators (e.g., STATUS_LOADING, STATUS_SUCCESS, STATUS_ERROR)
- LEVEL_: User level indicators (e.g., LEVEL_BEGINNER, LEVEL_INTERMEDIATE, LEVEL_EXPERT)

5.0 QUERYING

5.0 QUERYING

This section describes the query and retrieval capabilities of the system. The instructions necessary for recognition, preparation, and processing of a query applicable to a database shall be explained in detail. Use screen prints to depict examples of text under each heading.

5.1 Query Capabilities

Material History Query

- Purpose: Retrieve list of scanned materials for a specific user
- Parameters: User ID
- Output: List of materials with basic information and associated projects
- Location: History tab in main navigation

Query Process:

- User accesses History tab
- System queries user's scanned materials from backend API
- For each material, system retrieves material details and associated recycling projects
- Results are displayed in chronological order with project suggestions

Material Project Query

- Purpose: Find recycling projects that match scanned materials
- Parameters: Material name, material traits
- Output: Filtered list of relevant recycling projects
- Location: Material detail pages and history screen

Query Process:

- System receives material name and traits
- Queries recycling projects collection
- Filters projects based on material name matching and trait compatibility
- Returns up to 3 most relevant projects per material

5.2 Query Procedures

Material History Query Procedures

- User ID: Unique identifier for the user account
- Material ID: Identifier for specific material details

Sequenced Control Instructions:

5.0 Querying

- User Authentication
  - Verify user authentication token
  - Validate user permissions for data access
- Data Retrieval
  - Connect to backend API endpoint `/user/{userId}/materials`
  - Retrieve list of material IDs for the user
  - For each material ID, query `/material/{materialId}` for details
- Project Association
  - For each material, query `/recycling` endpoint
  - Filter projects based on material name matching
  - Apply trait-based filtering for project compatibility
- Result Processing
  - Format material data for display
  - Associate relevant projects with each material
  - Apply display limits (3 projects per material)
- Result Delivery
  - Package results for UI display
  - Return formatted response to client application

6.0 REPORTING

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This section describes and depicts all standard reports that can be generated by the system or internal to the user. Use screen prints as needed to depict examples of text under each heading.

6.1 Report Capabilities

No reporting capabilities are currently implemented in the Waste to Worth application.

6.2 Report Procedures

No report procedures are available as no reporting functionality is implemented in the Waste to Worth application.